ME4890: FINAL PROJECT

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Final group project overview:

- Several topics about nanosatellite are given. Your team needs to choose one and investigate the problem, search for resources, run simulation software if necessary, design the system for a specific mission, obtain results and data to support your design and method. You need to write a report and **present the work in the last week** of the term.
- different groups can choose the same topic.
- The report MUST be your own work. **Zero tolerance of plagiarism.** We will use software to check.
- The oral presentation will be arranged in the last week (i.e. 4th May). **The final project accounts for 40% in total; the report takes 20% and the presentation takes 20%.**
- You will assess the performance of your group mates by completing a peer assessment form after the presentation.

Format of report:

Word report
No page limit (suggest < 15 pages)
Include references.

Advice for writing a technical report:

1) Include essential sections such as abstract, introduction, your design and analysis, results… conclusion, reference.
2) Add a table of content at the beginning.
3) Please include all the details of your research, such as data, software simulation results, figures and tables. We'd like to see the evidence to support the design/conclusion. You can use Matlab, GMAT or any other software to produce results.
4) Include all the codes used in producing the results.

Marking of the report will be mainly based on the content/results/analysis.

Marking criteria

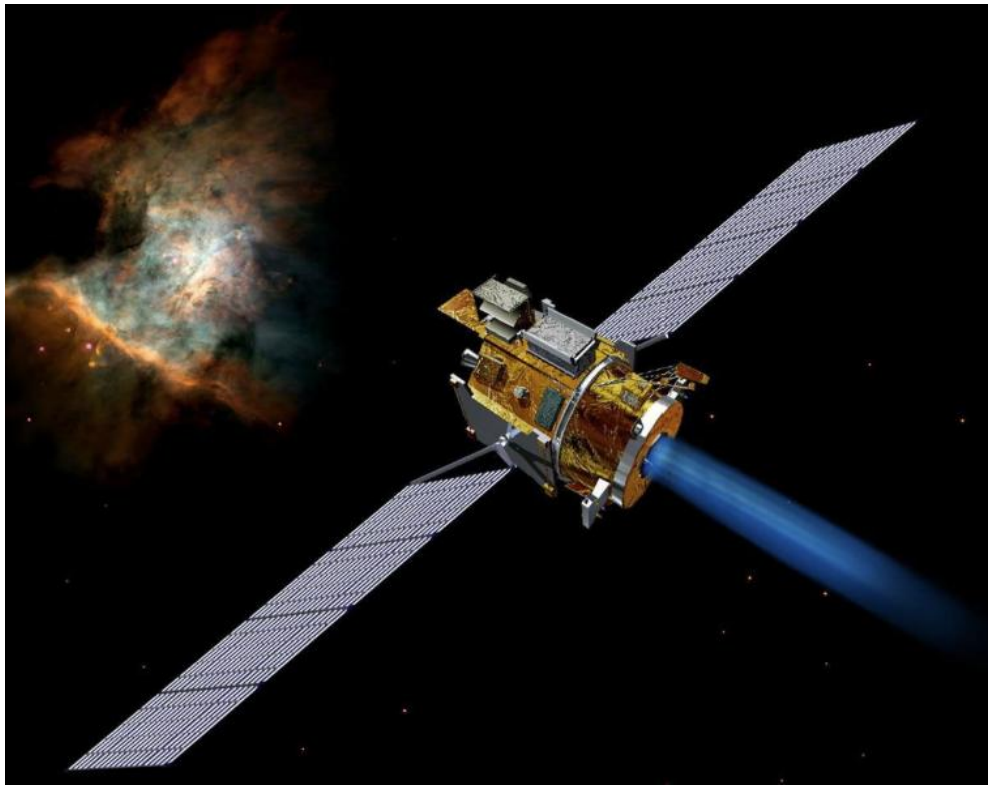
Marking criteria:

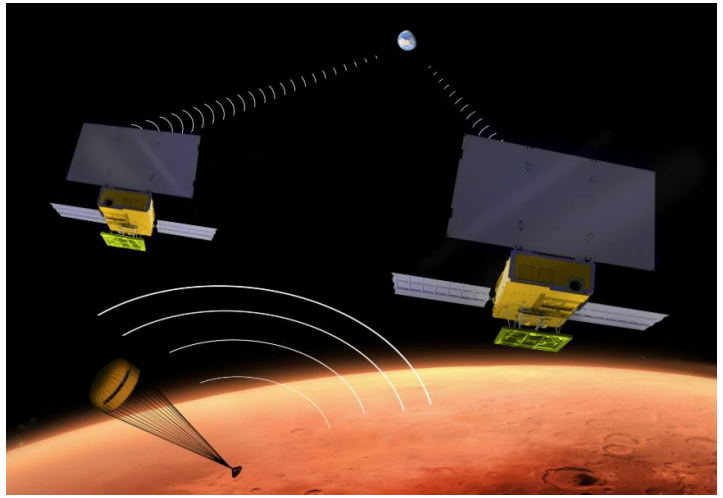
1. Completeness of the report (Is the overall structure complete?)
2. Organisation and logic of the report (Is the report well organised? How is the logic flow of writing?)
3. Understanding of the problem (Is the problem well explained? Is the student aware of the technical challenges?)
4. Feasibility of the design and the method (your explanation/reasoning/illustration, comparison study etc.)
5. Results and analysis (results produced from Matlab/GMAT/PSS simulations, data and figures which can support your conclusion).
6. Writing (Grammar etc.)

- We have several open topics for you to explore. If you are unsure about which specific direction to go, you are encouraged to contact the TA or me for help. We are happy to discuss with you about the ideas and technical details.

Final Project of ME4890

Topic 6: Cubesat for deep space exploration mission





Mission: Cubesat for deep space exploration mission

Major points to consider:

1. What is the primary mission objective for the deep space cubesat (e.g., remote sensing for space weather study in Mars/Jupiter, searching for evidence of water, communication relay, scientific study).
2. Cubesat orbit. What is the specific orbit for Mars or other planets? Can we draw the orbit and trajectories of the Cubesat? Estimation of the orbit period?
3. Cubesat design. Power system, battery capacity and solar panel design considering the space environment with reduced solar flux and different thermal space environment.
4. Propulsion submodule. What type of propulsion system will you select for the deep space task, e.g., electric, chemical, plasma or cold gas. And why do you choose it?
5. Communication with ground station on earth. How to design the communication link for deep-space distance, to ensure sufficiently large signal strength? Consider the data volume and data transferring efficiency.
6. Other things that you think are useful.